1. The larger rope is 4 m long and the smaller rope is 2 m long. This is expressed in the form of ratio as 4 : 2 and the simplest form of ratio of the larger rope to the smaller rope is 2 : 1 (See Fig. 3.1 (a))
2. The cost of a car is ₹ 5,00,000 and the cost of a motorbike is ₹ 50,000. This is expressed as 500000 : 50000 = 50 : 5 and the simplest form of ratio of the car to the motorbike is 10 : 1 (See Fig. 3.1 (b))

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20

3.2.3 Simplifying ratios of same unit

Example 3.1

Simplify the ratio 20 : 5.

Solution

Step 1 : Write the ratio in fraction form as $\frac{20}{5}$.

Step 2 : Divide numerator and denominator by 5. That is, $\frac{20 \div 5}{5 \div 5} = \frac{4}{1} = 4 : 1$

This is the ratio in the simplest form.

Example 3.2

Find the ratio of 500 g to 250 g.

Solution

500 g to 250 g = 500 : 250 $\Rightarrow \frac{500}{250} = \frac{500 \div 250}{250 \div 250} = \frac{2}{1} = 2 : 1$

This is the ratio in the simplest form.

Example 3.3

Madhavi and Anbu bought two tables for ₹ 750 and ₹ 900 respectively. What is the ratio of the prices of tables bought by Anbu and Madhavi?

Solution

The ratio of the price of tables bought by Anbu and Madhavi

$= 900:750 = \frac{900}{750} \Rightarrow \frac{900 \div 150}{750 \div 150} = \frac{6}{5} = 6:5$. This is the ratio in the simplest form.

3.2.4 Simplifying ratios of different units

1 hour = 60 minutes
20 x 1 = 20
20 x 2 = 40
20 x 3 = 60

Example 3.4

What is the ratio of 40 minutes to 1 hour?

Solution

Step 1 : Express the quantity in the same unit. (Hint : 1 Hour = 60 minutes)

Step 2 : Now, the ratio of 40 minutes to 60 minutes is $40:60 \Rightarrow \frac{40}{60} = \frac{40 \div 20}{60 \div 20} = \frac{2}{3} = 2:3$

This is the ratio in the simplest form.

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